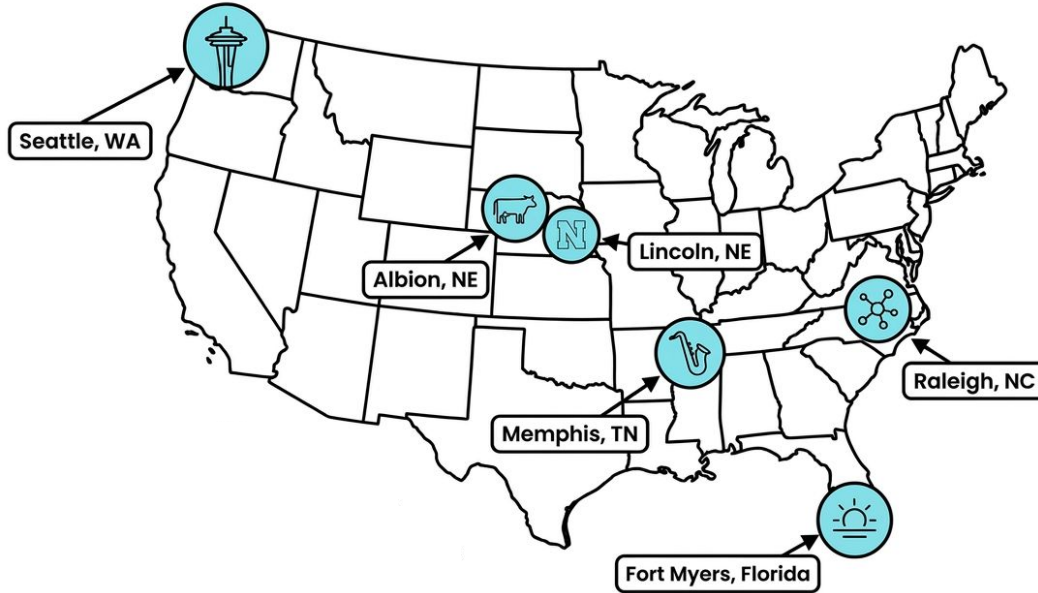


Graphs Are A Data Structure

Module 9 Week A — RDF, SPARQL, Fuseki

Get to know Lana — Fun Fact

AI.SPIRE

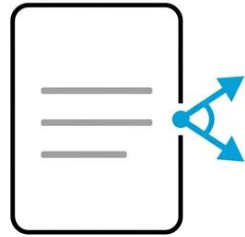


Places I've lived:

- Grew up in Albion, Nebraska
- College in Lincoln, Nebraska
- Lived in for work:
 - Memphis, Tennessee
 - Raleigh, North Carolina
 - Southwest Florida
- Moved to Seattle for marriage; great place for tech and startup

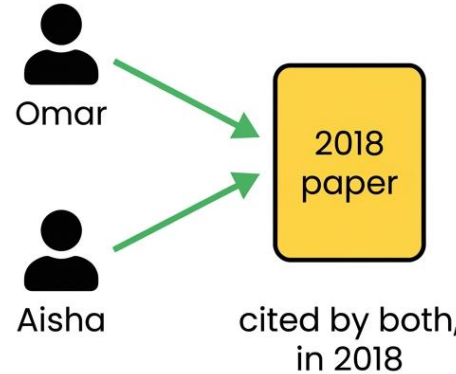
Two Questions

similarity question



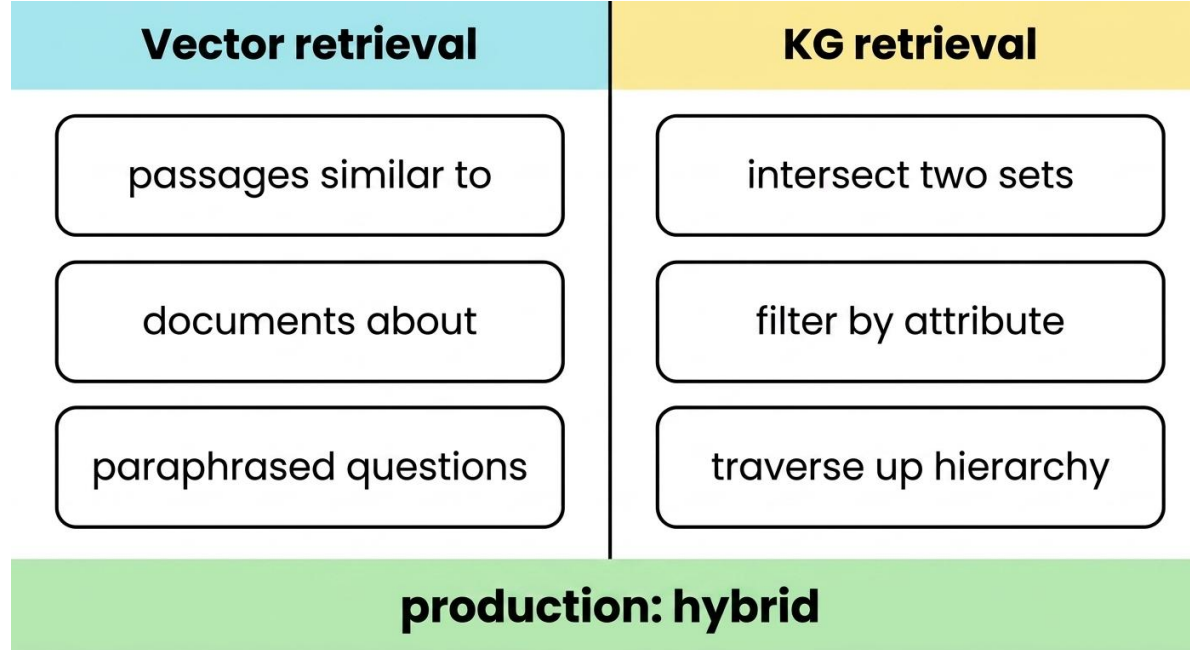
papers similar to this

graph traversal question



cited by both,
in 2018

Different Shapes Of Question



Subject Predicate Object

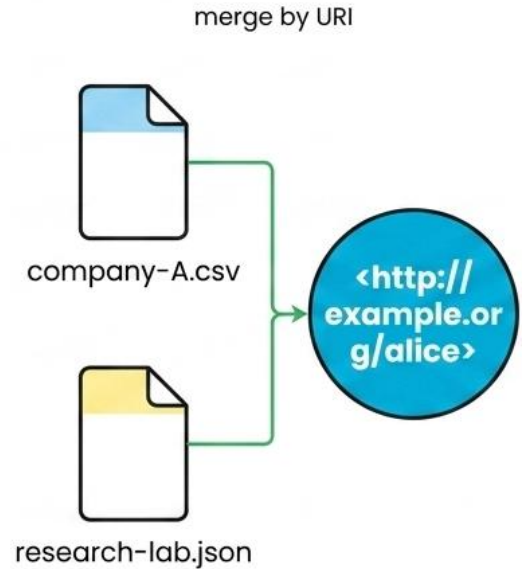


One RDF triple — queried with SPARQL

Names Are URIs



Expanded RDF triple composition



Prefixes Make Triples Shorter

expanded

<http://example.org/alice> <http://xmlns.com/foaf/0.1/knows> <http://example.org/bob> .

same triple

with prefix

**@prefix : <http://example.org/> .
@prefix foaf: <http://xmlns.com/foaf/0.1/> .
:alice foaf:knows :bob .**

Turtle Is The Text Format (Serialization)

Turtle

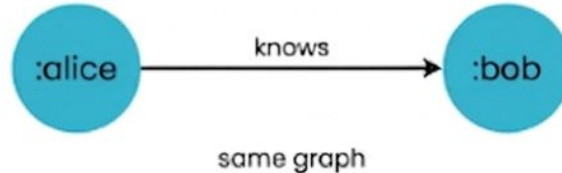
N-Triples

JSON-LD

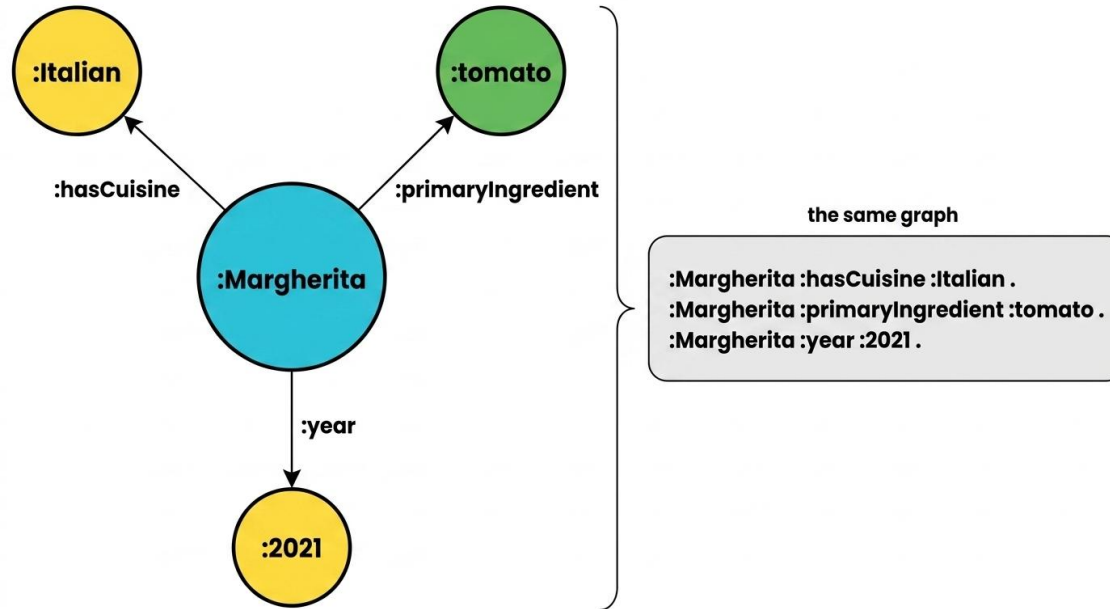
```
@prefix : <http://ex.org/> .
:alice :knows :bob .
```

```
<http://ex.org/alice> <http://ex.org/knows> <http://ex.org/bob> .
```

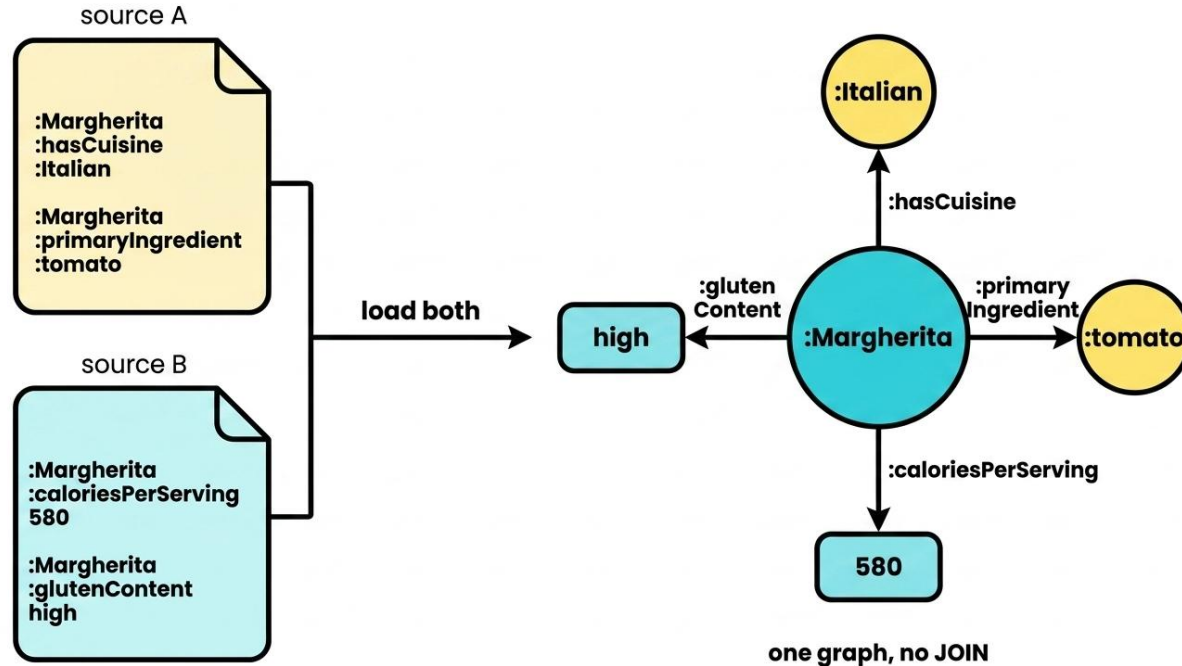
```
{
  "@id": "http://ex.org/alice",
  "http://ex.org/knows": {
    "@id": "http://ex.org/bob"
  }
}
```



A Four Node Graph



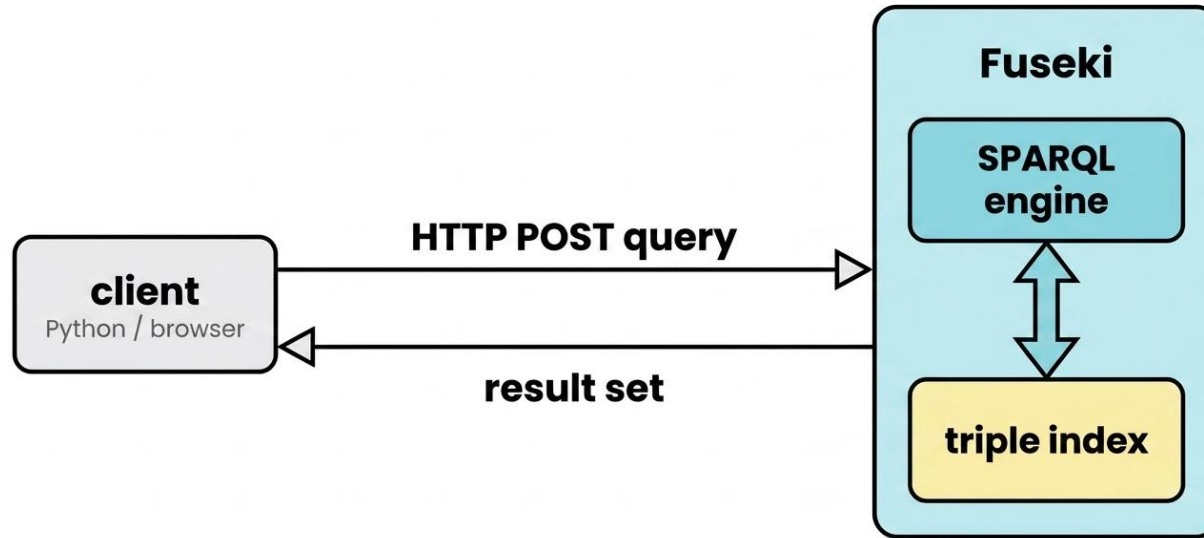
Two Sources, One Graph



When To Choose A Knowledge Graph

use a graph 	use a table 
shape varies	shape is fixed
many sources	one source
traversal query	aggregation query

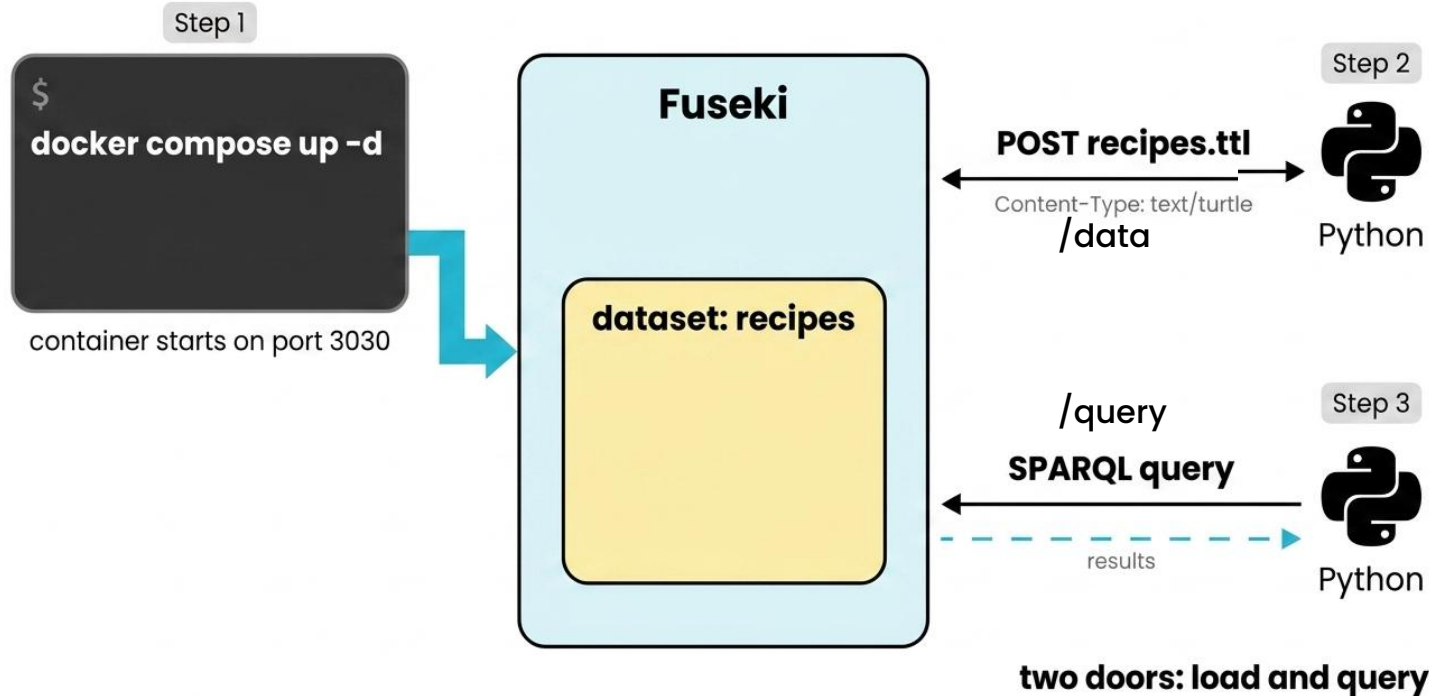
The Triple Store



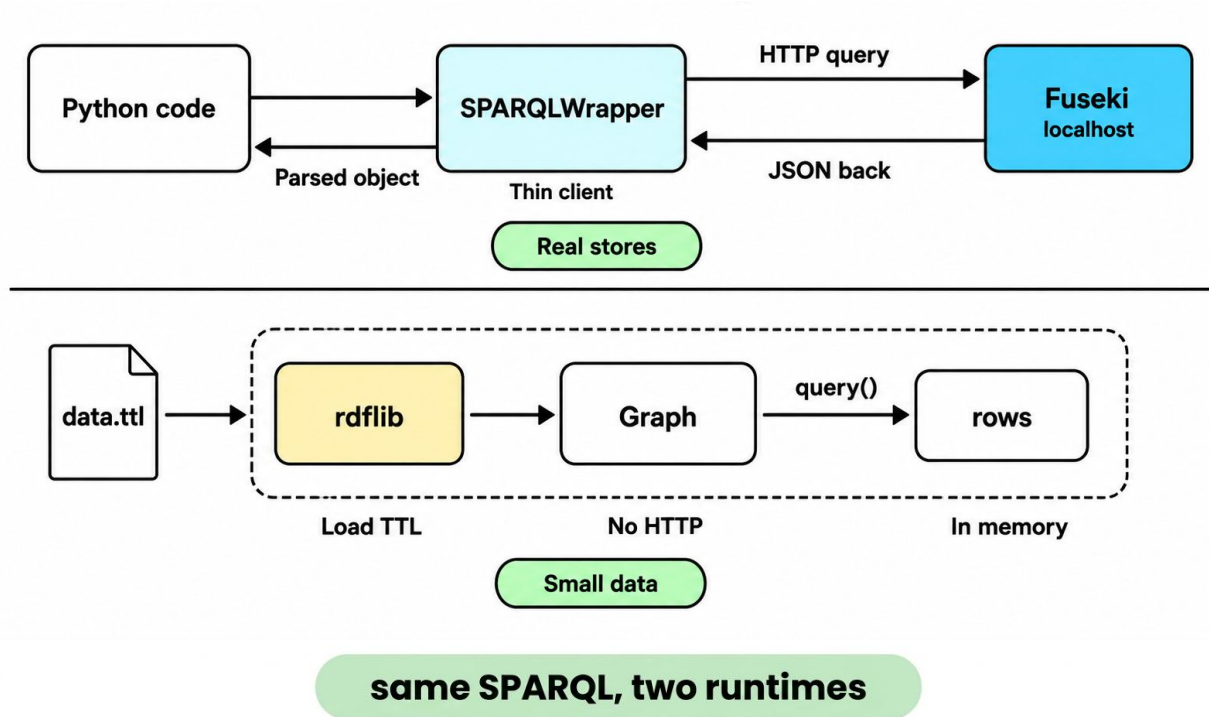
<http://localhost:3030/recipes/sparql>

the endpoint

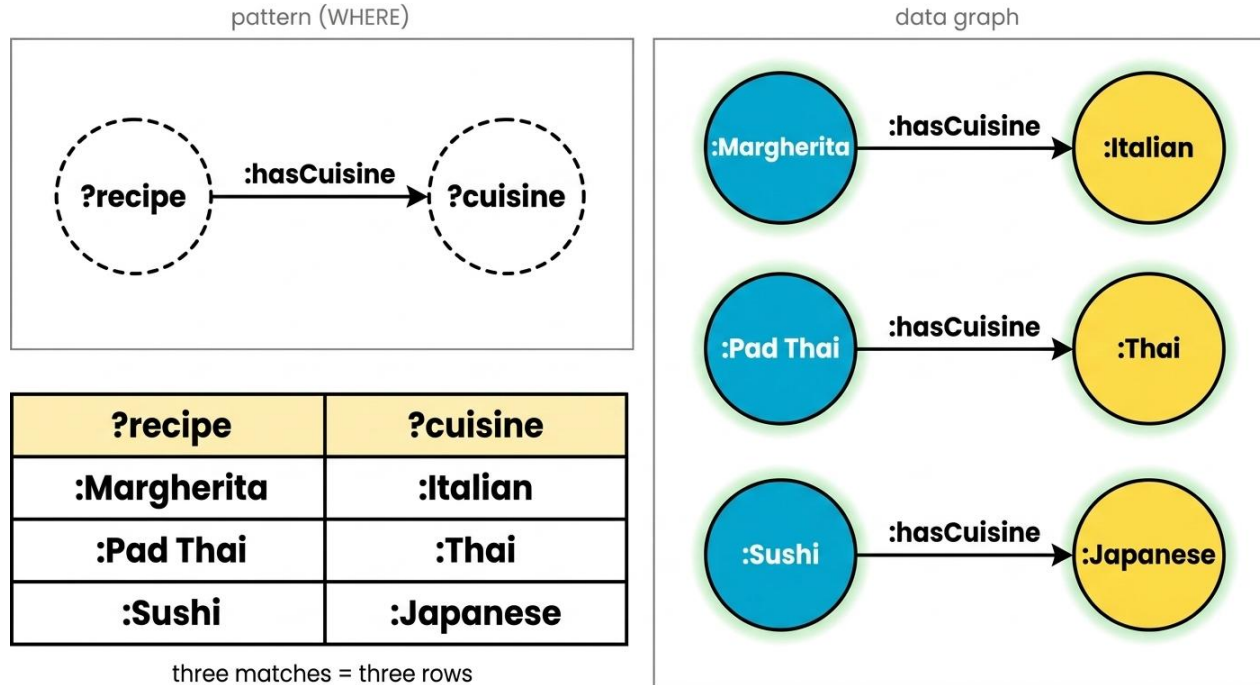
Loading A Dataset



Two Ways To Query RDF Data



SPARQL Is Pattern Matching



A First SELECT

PREFIX block

PREFIX : <http://ex.org/recipes/>

SELECT line

SELECT ?recipe ?cuisine

WHERE block

WHERE { ?recipe :hasCuisine ?cuisine . }

?recipe	?cuisine
:Margherita	:Italian
:Pad Thai	:Thai
:Sushi	:Japanese

Add A FILTER

PREFIX : <http://ex.org/recipes/>

```
SELECT ?recipe ?cuisine ?year
WHERE {
  ?recipe :hasCuisine ?cuisine .
  ?recipe :year ?year .
  FILTER (?year > 2020)
}
```

sieve

3 rows ↓ 1 row

?recipe	?cuisine	?year
:Margherita	:Italian	2021

OPTIONAL Makes Parts Optional 😊

```
SELECT ?recipe ?cuisine ?year
WHERE {
  ?recipe :hasCuisine ?cuisine .
  OPTIONAL { ?recipe :year ?year }
}
```

soft pattern

?recipe	?cuisine	?year
:Margherita	:Italian	2021
:Pad Thai	:Thai	—
:Sushi	:Japanese	2019

row kept, value missing

ORDER BY And LIMIT

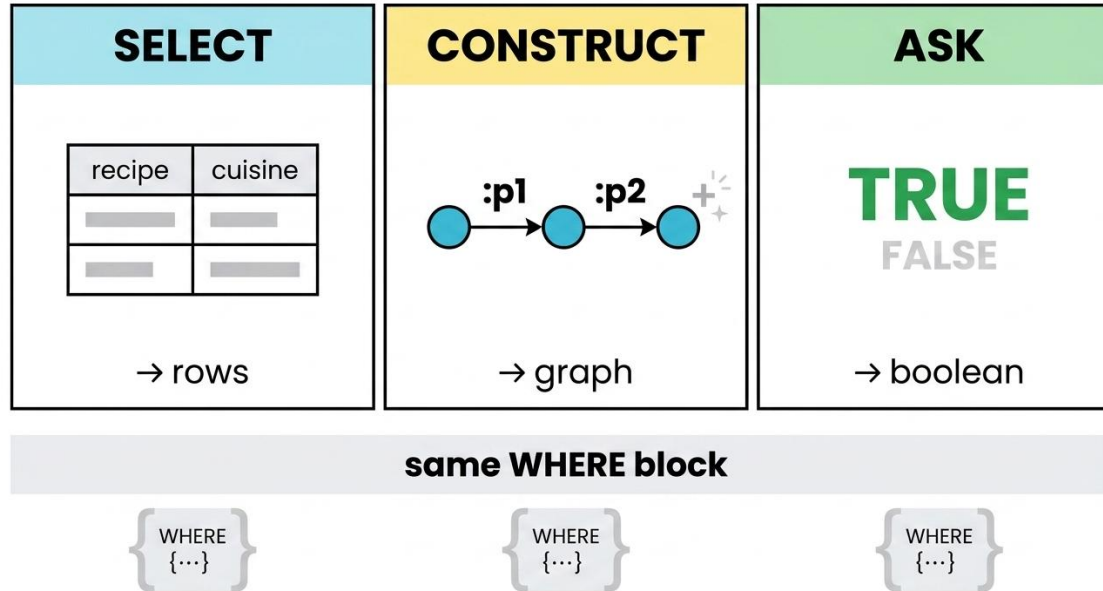
```
SELECT ?recipe ?year
WHERE {
  ?recipe :year ?year .
}
ORDER BY DESC(?year)
LIMIT 5
```

post-processing

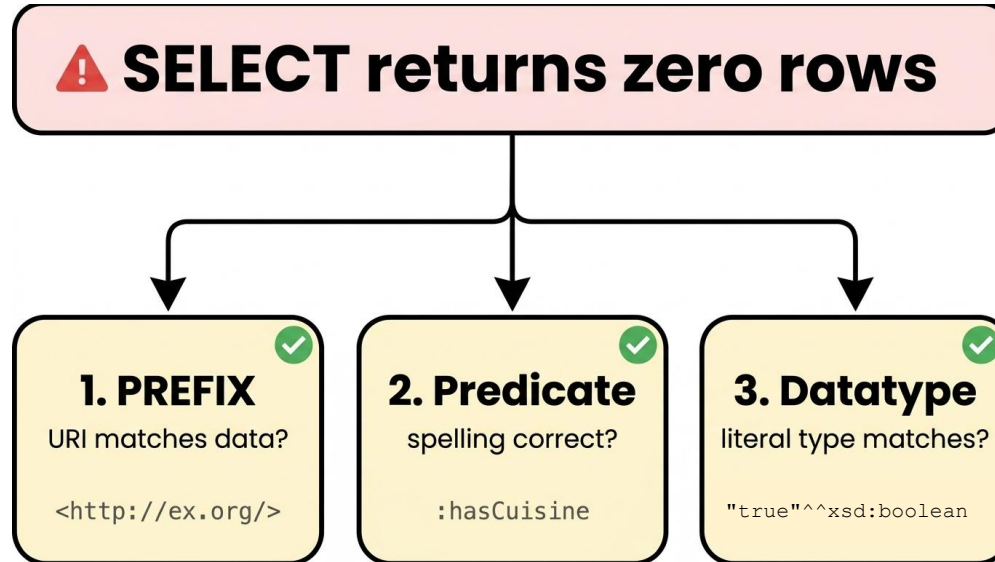
?recipe	?year ↓
:Margherita	2023
:Pad Thai	2022
:Sushi	2021
:Ramen	2020
:Tacos	2019

sorted, capped at 5

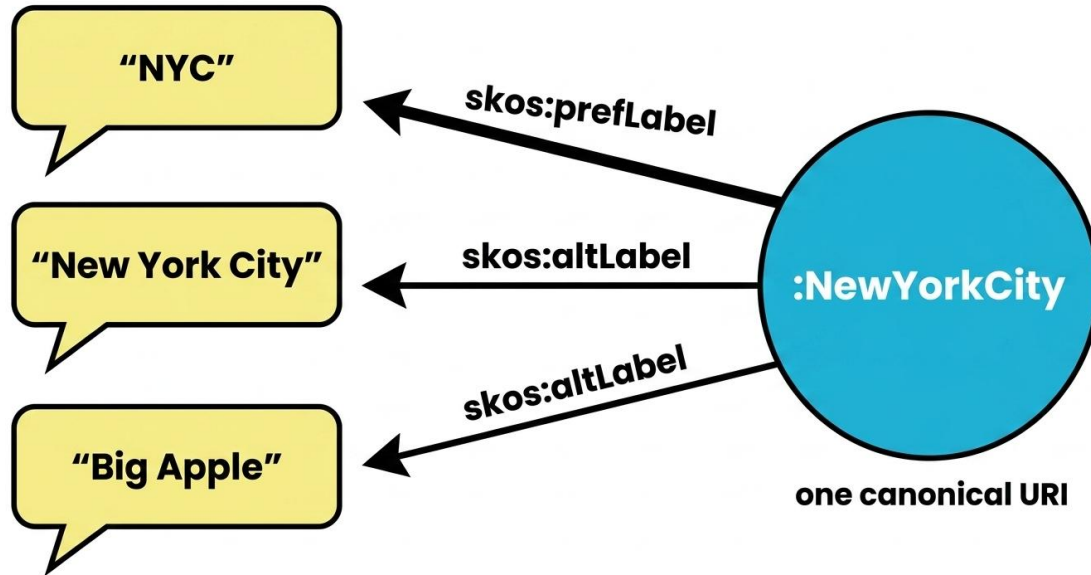
SELECT, CONSTRUCT, ASK



Zero Rows Returned



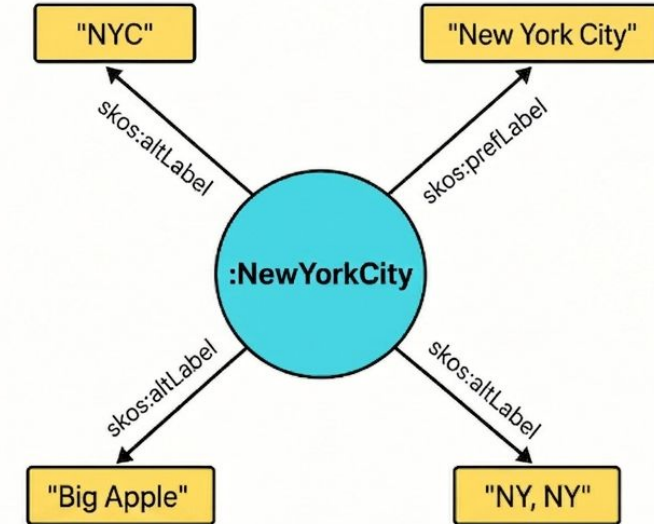
Strings Are Not Entities



SKOS In SPARQL

```
SELECT ?entity WHERE {
  VALUES ?label { "Big Apple" }
  { ?entity skos:prefLabel ?label }
  UNION
  { ?entity skos:altLabel ?label }
}
```

OR pattern



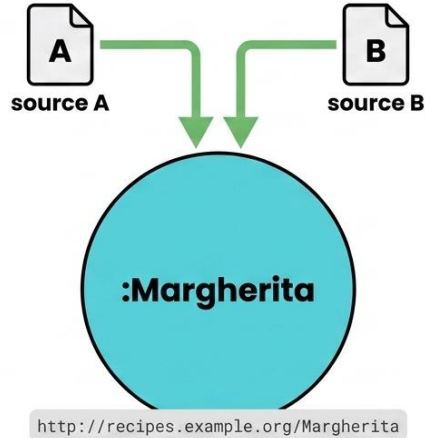
?entity

:NewYorkCity

→ canonical URI

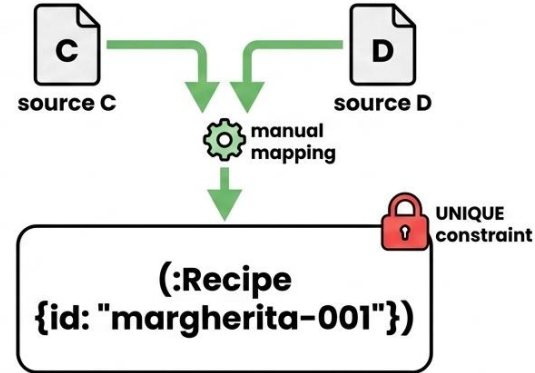
Identity Discipline

RDF / SPARQL



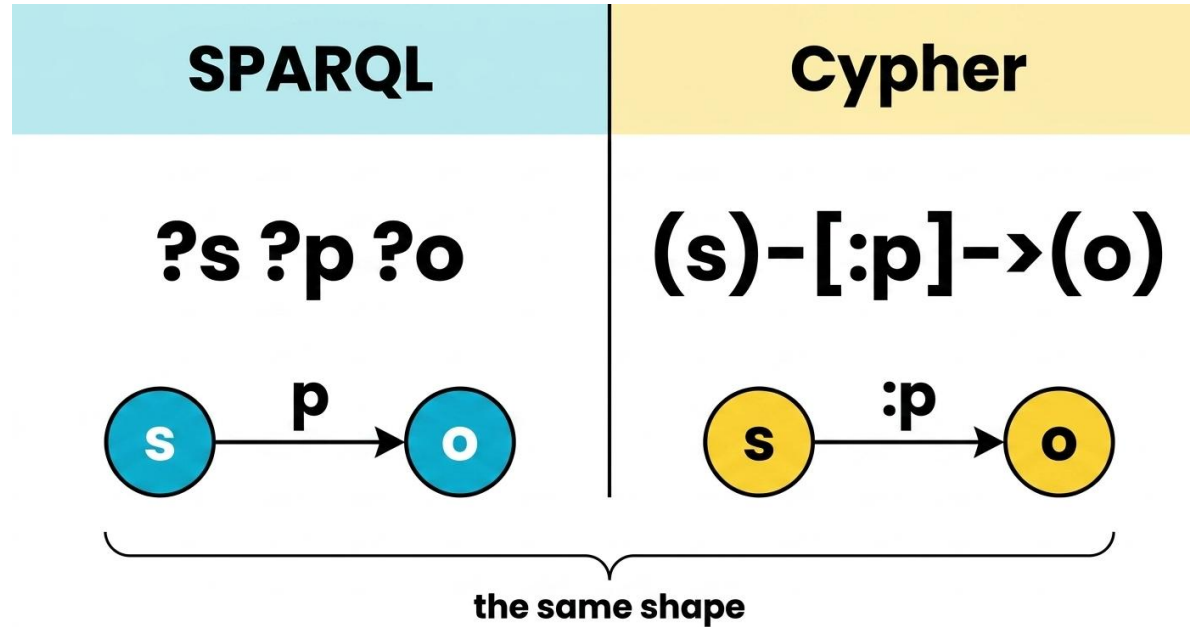
identity is automatic

Property graph / Cypher

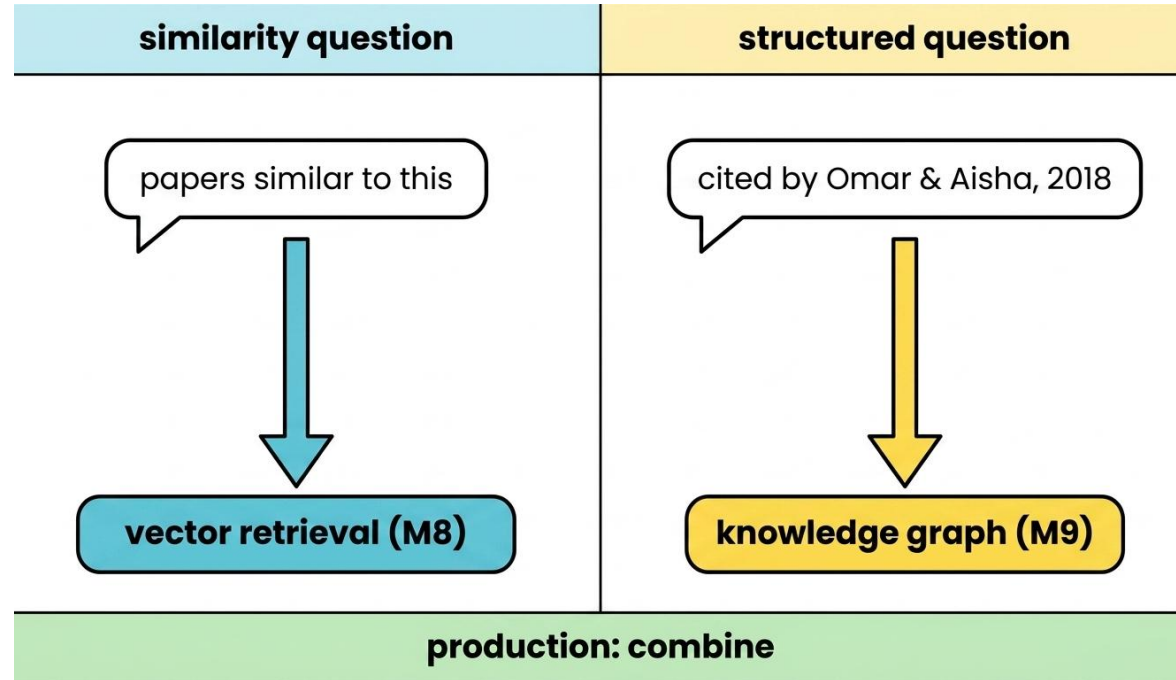


identity is declared

Same Shape, Two Notations



Back To The Two Questions



Questions

Announcements

Congratulations!



This week's Module Mentors

- Marwan
- Nayef
- Hosam